Hirmay Sandesara

Computational science, tensor networks, and quantum simulation

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Research Interests

My primary interest is in researching and developing **compressed algorithms and frameworks** for quantum many-body simulation. I am interested in memory-efficient representations, regimes where **tensor-network ranks remain manageable**, and scalable methods for fermionic dynamics and finite-temperature simulation.

Education

Technical University of Munich — Munich, Germany
M.Sc. Computational Science and Engineering, quantum computing focus — Oct 2024 – present

- Current grade: **1.5/5** (German scale; provisional, 22 October 2026).
- **TUM merit-based tuition waiver**; **Deutschlandstipendium**; CSE student representative.

Ahmedabad University — Ahmedabad, India
B.Tech. Computer Science and Engineering — Jul 2019 – May 2023

- Grade: 1.4 (German equivalent; 3.73/4.0); ranked **2nd of 140 students**.
- Dean's Recognition List; graduated with a certificate of highest excellence.

Selected Publications

H. Sandesara, B. Decker, and C. Mendl. "MPS-Compressed Majorana Propagation with Virtual $\mathbb{Z}_2$ Charges." Manuscript in preparation.

H. Sandesara, A. Shukla, and P. Vedula. "A quantum approach for optimal control." *Quantum Information Processing* **24**, 95 (2025). doi:10.1007/s11128-025-04710-z.

H. Sandesara et al. "Space Sustainability Using Quantum Computing." White paper, OSF Preprints (2022). doi:10.31219/osf.io/wndfb.

Research Experience

Technical University of Munich — Munich, Germany
Master's thesis in tensor-network algorithms; supervisor: Prof. Christian B. Mendl — Current

- Developing an **operator-space matrix product state (MPS)** approach for propagating Majorana operators in fermionic systems with long-range interactions.
- Introducing **virtual $\mathbb{Z}_2$ charge labels** on the MPS bonds to record accumulated fermionic parity, with the local rule $q_R = q_L \oplus p(s)$.
- Handling signs from long Jordan–Wigner and Majorana strings through endpoint and virtual-state information while preserving charge-aware tensor contractions and **SVD compression**.

Fraunhofer Institute for Integrated Circuits IIS
Student Researcher, Quantum Compilation Group — May 2026 – Oct 2026
Supervisors: Sarah Volkamer and Daniel D. Scherer.

- Integrated existing **automated cut-selection code** into a circuit-cutting software package under development, adapting the implementation for use within the package workflow.
- Surveyed compilation and cut-selection methods for quantum circuit cutting, including multi-qubit gate cutting, state-of-the-art approaches, open research questions, and relevant concepts from ZX calculus; prepared selected findings for internal technical discussions.
- Created Jupyter notebooks to demonstrate and test selected cut-finding approaches and their integration with the circuit-cutting package.

Ahmedabad University — Ahmedabad, India
Research Assistant — Sep 2023 – May 2024

- Developed a multidimensional formulation for nonlinear optimal-control problems using constrained Hamiltonian dynamics and canonical quantization.
- Designed and implemented a **variational quantum method** for the resulting broken-PT-symmetric, non-Hermitian systems using Python and Mathematica.
- Evaluated control problems with PennyLane, Qibo, SciPy, symbolic computation, parallel integration, and PDE solvers; work published in *Quantum Information Processing*.

Quantum Open Source Foundation Remote
Mentee; mentor: Ryan Hill, qBraid Apr 2024 – Aug 2024

- Selected as one of approximately 40 participants from more than 200 applicants.
- Worked on the design of a Python framework for configurable quantum reservoir-computing models and integration with QuEra computing resources.

Artificial Brain Remote
Quantum Computing Intern Jul 2022 – Dec 2022

- Co-authored a white paper formulating space-debris removal as knapsack and routing optimization problems and surveying QAOA and quantum-annealing approaches.
- Contributed to quantum scheduling for Earth-observation satellites and to a D-Wave portfolio-optimization prototype [🌐] targeting the Sharpe ratio.

Selected Technical Projects

Entanglement distillation on a constrained quantum network 2026
MIT iQuHACK, IonQ Challenge – 2nd prize

- Led a four-person team that designed a heuristic parity-check protocol for entanglement distillation under a minimal single-Bell-pair budget.
- Achieved a challenge score of 89 and claimed network edges at fidelity at least 0.9; see the code and technical report.

NISQ methods for open-system simulation 2024
QHack Open Hackathon [🌐] – 3rd prize, Seeing the Future; AWS Top 3

- Led development of shallow parameterized circuits for state preparation and time evolution in open quantum systems.
- Tested the state-preparation routine on synthetic inputs and Alzheimer's patient data using **parallel GPU optimization** with NVIDIA cuQuantum; also explored logical cat-qubit simulations. Code and report.

Hamiltonian simulation of coupled classical oscillators 2024
Classiq QRISE Challenge – winning submission

- Derived the Hamiltonian and implemented state preparation and simulation workflows for a coupled-oscillator model on Classiq's platform.
- Compared construction choices with the practical aim of maintaining fidelity while reducing circuit depth. Project repository.

Honors and Awards

2nd Prize, MIT iQuHACK IonQ Challenge 2026
3rd Prize, QHack Seeing the Future Challenge; AWS Top 3 2024
Winner, QRISE Classiq Challenge 2024
TUM merit-based tuition waiver and **Deutschlandstipendium** 2024
Rank 2/140, Ahmedabad University; Dean's Recognition List 2023

Open Source, Mentoring, and Service

- Contributed an implementation of uniform-superposition state preparation to Qiskit's circuit library; co-authored material on Grover's algorithm for the Qiskit Textbook; and helped fix a Qiskit Terra bug.
- Translated more than 5,000 words of Qiskit educational material into Hindi.
- Mentored participants during the IBM Quantum Challenges in 2022 and 2024 by debugging tasks and explaining relevant concepts.
- Organized quantum-computing workshops and hackathons.

Technical Skills

Numerical methods	Matrix product states/operators, tensor contractions, SVD and truncation, numerical linear algebra, optimization, PDE solvers, quantum simulation
Programming	Python, Julia, C, C++, MATLAB, Mathematica, SQL
Libraries and tools	NumPy, SciPy, SymPy, Qiskit, PennyLane, Qibo, QuTiP, cuQuantum, Matplotlib, PyTest, Git, LaTeX